

Modular Arithmetic, Residues & Power Cycles - Assimilation Book

1. RECONNECT - from divisibility to congruence

From prior divisibility work, $m|(a-b)$ means a and b differ by a multiple of m . Modular arithmetic compresses that into $a \text{ congruent } b \pmod{m}$.

It means same remainder modulo m , not ordinary equality.

Example: 17 congruent 5 (mod 12) because $12|(17-5)$.

Contrast: equality vs congruence

17 is not equal to 5, but they are congruent modulo 12. Equality is exact; congruence depends on a chosen modulus.

2. DISCOVER - residues are states

Modulo 7 every integer belongs to one of seven states: 0,1,2,3,4,5,6. You may replace a number by any congruent representative when using legal operations.

If $a \text{ congruent } b \pmod{m}$ and $c \text{ congruent } d \pmod{m}$, then:

- $a+c \text{ congruent } b+d$;
- $a-c \text{ congruent } b-d$;
- $ac \text{ congruent } bd$;
- positive integer powers preserve congruence.

So reduce early: $1234567 \text{ mod } 9$ may be replaced by its residue before further modular arithmetic.

3. MAKE SENSE - cancellation is conditional

From $ac \text{ congruent } bc \pmod{m}$, cancellation of c is safe modulo m when $\gcd(c,m)=1$, because c then has a modular inverse.

Counterexample: $2 \cdot 1 \text{ congruent } 2 \cdot 4 \pmod{6}$ since $2 \text{ congruent } 8 \pmod{6}$, but 1 is not congruent to 4 (mod 6). The cancelled factor 2 is not invertible mod 6.

If $3x \text{ congruent } 6 \pmod{7}$, inverse of 3 mod 7 is 5, so $x \text{ congruent } 30 \text{ congruent } 2 \pmod{7}$.

Never write modular division before asking: is the divisor invertible modulo m ?

4. Power cycles

Compute only enough powers to see repetition.

Modulo 7:

$2^1, 2^2, 2^3, \dots$ gives residues 2,4,1,2,4,1,....

So $2^{100} \text{ mod } 7$ uses $100 \text{ mod } 3 = 1$, hence residue 2.

For last digits, choose the modulus from the target:

- last digit $\rightarrow \text{mod } 10$;
- last two digits $\rightarrow \text{mod } 100$.

The 2024 anchor 5^{2024} stabilizes at 25 modulo 100 after exponent 2.

5. Base cycle and exponent cycle can both matter

For $n^n \text{ mod } 7$, the base depends on $n \text{ mod } 7$; for nonzero residues, the exponent can be reduced through a period dividing 6. A universal period must preserve both pieces of information. That produces the 2025 anchor period $\text{lcm}(7,6)=42$.

Do not blindly reduce the exponent modulo 6 when the base is divisible by 7; handle zero-residue cases explicitly.

6. Simultaneous congruences

Solve

$x \text{ congruent } 2 \pmod{3}$ and $x \text{ congruent } 3 \pmod{5}$.

Numbers in the first class are 2,5,8,11,...; 8 is the first also congruent to 3 mod 5. Solutions repeat every 15:

$x \text{ congruent } 8 \pmod{15}$.

With non-coprime moduli, consistency matters. Example:

$x \text{ congruent } 1 \pmod{4}$, $x \text{ congruent } 3 \pmod{6}$ has solution $x \text{ congruent } 9 \pmod{12}$.

But $x \text{ congruent } 1 \pmod{4}$, $x \text{ congruent } 2 \pmod{6}$ is impossible because the requested residues disagree modulo $\gcd(4,6)=2$.

At Grade-9 depth, listing or parametrizing one congruence and checking the other is often cheaper than quoting a general theorem.

7. TRY - attempt before help

Track A - meaning

Show 83 congruent 11 (mod 12).

- Full support: subtract and show divisibility by 12.
- Medium support: same remainder means difference divisible by modulus.
- Light support: compare remainders.
- Independent: decide whether 137 congruent 5 (mod 11).

Track B - cycle

Find $3^{100} \bmod 7$.

- Full support: list residues of $3, 3^2, \dots$ until 1 returns; use exponent modulo cycle length.
- Medium support: find the cycle before touching exponent 100.
- Light support: huge power $\rightarrow$ small repeated residue state.
- Independent: find the last digit of 7^{2026} .

Track C - cancellation

Solve $4x$ congruent 3 (mod 7).

- Full support: inverse of 4 mod 7 is 2; multiply both sides.
- Medium support: test $\gcd(4, 7) = 1$, then invert.
- Light support: cancellation/division requires invertibility.
- Independent: analyze $6x$ congruent 9 (mod 15) without illegal cancellation.

Track D - simultaneous congruences

Solve x congruent 1 (mod 4), x congruent 4 (mod 5).

- Full support: write $x = 1 + 4k$ and test modulo 5.
- Medium support: parametrize one residue class.
- Light support: combine two repeating lists.
- Independent: solve x congruent 2 (mod 6), x congruent 5 (mod 9) or prove impossible.

8. DIAGNOSE

- Equality reflex: congruent numbers need not be equal.
- Divisibility reflex: a congruent b is a statement about $a-b$, not only about either number separately.
- Expansion reflex: large powers usually call for a cycle.
- Cancellation reflex: common factor does not automatically cancel modulo m .
- Wrong modulus: last two digits are mod 100, not mod 10.
- CRT reflex: simultaneous congruences may be incompatible when moduli share factors.

9. ADOPT - six mental rules

1. a congruent b (mod m) means $m \mid (a-b)$.
2. Reduce residues early under legal operations.
3. Addition, subtraction and multiplication are safe; division is conditional.
4. A factor is invertible mod m exactly when its $\gcd$ with m is 1.
5. Huge powers $\rightarrow$ search for a cycle before expanding.
6. Multiple congruences $\rightarrow$ check compatibility, then find the repeating combined class.

10. TRANSFER

Residues later become:

- last-digit and place-value filters in later number-theory work;
- parity/residue invariants in later combinatorial work;
- finite states in cyclic processes;
- collision detectors: two values share a residue iff their difference is divisible by the modulus.

Modular Arithmetic, Residues & Power Cycles - First-Step Reference

Recognition atlas

Clue | First move

same remainder / congruent | write modulus divides the difference

huge power modulo m | reduce base; list a short power cycle

last digit | work mod 10

last two digits | work mod 100

modular equation with a coefficient | test whether the coefficient is invertible

two congruences | parametrize one class, then test the other

shared-factor moduli | check compatibility modulo their gcd first

Legal-operation checklist

From a congruent $b \pmod{m}$ you may safely add, subtract, multiply and take positive integer powers. Division or cancellation needs an invertible factor.

Before cancelling c from $ac \pmod{m}$, check $\gcd(c,m)=1$.

Power-cycle checklist

1. Choose the modulus from the target.
2. Reduce the base.
3. List powers only until the residue state repeats.
4. Reduce the exponent by the justified cycle length.
5. Handle zero/non-invertible residue cases separately when needed.

Simultaneous-congruence checklist

1. Check compatibility if moduli share factors.
2. Write one congruence as a parametrized class.
3. Substitute into the other congruence.
4. State the combined repeating class and its period.

WHY-NOT reminders

- Congruence is not ordinary equality.
- A common factor does not always cancel.
- A large exponent is not a signal to expand.
- The target determines the modulus; do not default to mod 10.

Modular Arithmetic, Residues & Power Cycles - Recognition and First-Line Lab

Write only the first useful line unless asked for a value.

1. Show $38 \equiv 3 \pmod{5}$ using divisibility.
2. Decide whether $17=5$ and whether $17 \equiv 5 \pmod{12}$.
3. Reduce 12345 modulo 7 before any further work.
4. Write the first four residues of $2^n \pmod{7}$.
5. Last digit of 3^{100} : state the modulus first.
6. Last two digits of 7^{50} : state the modulus first.
7. From $3x \equiv 6 \pmod{7}$, write the inverse step.
8. From $2x \equiv 4 \pmod{6}$, state why direct cancellation modulo 6 is unsafe.
9. Solve-first-line: $x \equiv 2 \pmod{3}$, $x \equiv 1 \pmod{4}$.
10. Compatibility check: $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{6}$.
11. Distinct residues of A and B modulo m: write the collision condition to avoid.
12. For $n^n \pmod{7}$, name the two periodic pieces that must both be preserved.
13. Explain $a \equiv b \pmod{m}$ in one sentence without congruence notation.
14. Find an inverse candidate for 5 modulo 12.
15. State whether 6 has an inverse modulo 15 and why.
16. If $a \equiv b \pmod{9}$, write one valid statement involving a^2, b^2 .

Modular Arithmetic, Residues & Power Cycles - Practice and Transfer Bank

All items are author-created.

F0 Foundation

1. Find residues of $-1, 8, 15, 22$ modulo 7.
2. Verify $35 \equiv 5 \pmod{10}$ from a difference.
3. Reduce 123456 modulo 9.
4. List the residue classes modulo 5.

F1 Direct

5. Find $2^{40} \pmod{7}$.
6. Find the last digit of 7^{2026} .
7. Find the last two digits of 5^{37} .
8. Solve $3x \equiv 5 \pmod{7}$.
9. Solve $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$.

F2 Standard

10. Find $3^{100} \pmod{7}$.
11. Find the last digit of $2^{2025} + 3^{2025}$.
12. Solve $5x \equiv 8 \pmod{12}$.
13. Analyze $4x \equiv 8 \pmod{12}$ and list all residue classes x modulo 12.
14. Determine whether $x \equiv 1 \pmod{6}$, $x \equiv 4 \pmod{9}$ is compatible.
15. Find the least positive x with $x \equiv 3 \pmod{4}$, $x \equiv 5 \pmod{7}$.

F3 Disguised

16. Find the last two digits of 3^{20} .
17. Prove every power 5^k with $k \geq 2$ ends in 25.
18. Find the period of $2^n \pmod{7}$.
19. Find the period of $3^n \pmod{7}$.
20. If $a \equiv b \pmod{11}$, prove $a^3 + 2a \equiv b^3 + 2b \pmod{11}$.
21. Give a counterexample to unrestricted cancellation modulo 8.
22. Solve $6x \equiv 9 \pmod{15}$ as residue classes modulo 15.

F4 Preliminary-style

23. Find $2^{1000} + 3^{1000} \pmod{7}$.
24. Find the last two digits of 11^{2026} .
25. Find the least positive x satisfying $x \equiv 4 \pmod{6}$ and $x \equiv 7 \pmod{9}$, or prove none exists.
26. Determine all residues $x \pmod{21}$ satisfying $7x \equiv 14 \pmod{21}$.
27. Find the smallest modulus $m \geq 14$ for which $1^4, \dots, 14^4$ are all distinct residues.
28. Explain why a universal period of $n^n \pmod{7}$ must be divisible by both 7 and 6.
29. WHY-NOT: from $10x \equiv 20 \pmod{30}$, why may you not conclude $x \equiv 2 \pmod{30}$?
30. A clock state satisfies $t \equiv 2 \pmod{4}$ and $t \equiv 5 \pmod{6}$. Decide whether such a state exists.

Modular Arithmetic, Residues & Power Cycles - Independent Mixed Mastery Check

No default hints or method labels.

Part A - Notice / first line

1. $83 \equiv 11 \pmod{12}$: write the divisibility statement underneath it.
2. Find the first line for $7^{123} \pmod{10}$.
3. $4x \equiv 5 \pmod{9}$: write the legality check before solving.
4. $x \equiv 2 \pmod{6}$, $x \equiv 5 \pmod{9}$: write the compatibility check.

Part B - Full mixed solve

5. Find $2^{2026} \pmod{7}$.
6. Find the last two digits of 5^{2026} .
7. Solve $5x \equiv 7 \pmod{12}$.
8. Solve $6x \equiv 12 \pmod{18}$ as classes modulo 18.
9. Find the least positive x with $x \equiv 1 \pmod{4}$, $x \equiv 4 \pmod{5}$.
10. Find $3^{1234} \pmod{7}$.
11. Determine whether $x \equiv 2 \pmod{4}$, $x \equiv 3 \pmod{6}$ has a solution.
12. Find the last digit of 7^{7^7} .

Part C - Same surface, different decision

13. Compare $14|42$ with $42 \equiv 0 \pmod{14}$ and explain the relationship without calling the statements identical in form.
14. Compare cancellation in $3x \equiv 6 \pmod{7}$ and $3x \equiv 6 \pmod{9}$; solve both.

Part D - Transfer / WHY-NOT

15. Find the smallest $m \geq 14$ for which the residues of $1^4, \dots, 14^4$ are distinct.
16. A learner expands 2^{100} to find its residue mod 7. Give the shorter structural route and explain why it is valid.